GenAI for Engineers Lesson Plan

IES | Delivery plan aligned to the supplied revised v2 proposal and lesson plan

Deliver one lab-driven day for engineers and officers across disciplines. Preserve the six modules, Primer 3.0, all 16 named labs and the seven learning outcomes. Minimum class size in the supplied plan is 20 learners. Use pairs for peer review while each learner retains their own evidence.

The day runs 09:00-17:00. There are 390 instructional minutes, 60 lunch minutes and 30 tea-break minutes. The source phrase seven training hours excluding lunch includes the two tea breaks. Lab allocations total 300 minutes, approximately 77 percent of instruction. Embedded demonstrations and peer review stay inside the specified lab allocations; do not add extra time to Module 4.

Petroleum delivery option

Petroleum transfer pathway: choose this scenario instead of the original lab within the same time allocation. Additional explanations are reference reading. All petroleum records, limits, outputs and scores here are original fictional teaching examples; no operating action is authorised. Choose the scenario before class; do not require both variants. Lab 3.2 transfers the DMAIC/SPC method to synthetic pressure readings. Lab 4.4 uses the forecast and event-evaluation examples. Preserve the original methodology primer and core checks. The 27 added slides are a reference bank, not extra lecture time.

Prerequisites and trainer resources

Corporate Microsoft 365 account and active Copilot licence; Word, Excel, PowerPoint, Teams and SharePoint; enabled agent creation and reasoning agents; own laptop. Prepare an approved training SharePoint site, current and obsolete synthetic references, two test accounts with different permissions, the editable slide deck, Learner Guide, lab folders, methodology one-pagers, FMEA scale and document templates. No previous methodology knowledge is assumed.

Learning outcomes

LO1 Explain FMEA, Six Sigma DMAIC and SPC, their uses, terms and outputs before applying them.

LO2 Create and refine engineering documents using Copilot in Word, Excel and PowerPoint.

LO3 Use R.I.S.E. and E-MAP with iterative prompting to obtain verifiable engineering outputs.

LO4 Build, test and share agents for FMEA, reporting and multi-step engineering workflows.

LO5 Use Researcher and Analyst for cited research and engineering data analysis.

LO6 Ground engineering answers on SharePoint sources with citations and permission checks.

LO7 Apply all seven steps of the AI Security Framework to engineering deliverables.

Detailed delivery schedule

Start	End	Content and activity	Module
09:00	09:10	Welcome, scope and ecosystem framing	1
09:10	09:20	Lab 1.1 onboarding	1
09:20	09:40	Case walkthrough, tool choice and evidence	1
09:40	09:55	Lab 1.2 challenge	1
09:55	10:00	Review-inclusive timing debrief	1
10:00	10:15	RISE, E-MAP and source authority	2
10:15	10:35	Lab 2.1 specification	2
10:35	10:50	Lab 2.2 inspection report	2
10:50	11:10	Lab 2.3 risk and method statement	2
11:10	11:15	Evidence and release-gate debrief	2
11:15	11:30	Tea break	-
11:30	11:45	Primer 3.0 FMEA 5 min, DMAIC 5 min, SPC 4 min, orientation 1 min	3
11:45	12:15	Lab 3.1 FMEA	3
12:15	12:40	Lab 3.2 DMAIC and SPC	3
12:40	13:00	Lab 3.3 spec pack with embedded mini challenge	3
13:00	14:00	Lunch	-
14:00	14:25	Lab 4.1 FMEA Assistant including demonstration	4
14:25	14:45	Lab 4.2 Report Writer including demonstration	4
14:45	15:00	Lab 4.3 recurring workflow chain	4
15:00	15:15	Lab 4.4 Researcher and Analyst	4
15:15	15:30	Tea break	-
15:30	15:35	SharePoint scope and governance demonstration	5
15:35	16:00	Lab 5.1 knowledge agent	5
16:00	16:15	Lab 5.2 end-to-end handoff	5
16:15	16:35	Lab 6.1 seven-step security review	6
16:35	16:45	Lab 6.2 personal toolkit	6
16:45	17:00	Closing, shared prompts, next actions and Q&A	All

Module 1 The Copilot ecosystem for engineers

09:00-10:00 | 60 minutes | LO2, LO3

Slide reference: 1-19. Use the deck selectively for explanation and as a lab reference; do not lecture through every slide.

From language generation to engineering evidence

A language model composes a response from the context supplied to it. Its fluency does not establish that an equipment condition, calculation or source is correct. In this course, Copilot is used to structure evidence and draft documents while the engineer retains responsibility for interpretation and release. PS-17 has an intermittent indicated-flow event, but the source does not establish a failed pump. This distinction is the central review habit throughout the day.

Choosing the working surface

Use Copilot Chat to structure a short source text or question. Use Word to develop a document, Excel to inspect data and calculations, and PowerPoint to communicate the reviewed conclusion. Agents package instructions and selected knowledge for repeated tasks. SharePoint provides a controlled source location. These roles complement each other; a chat session does not automatically create a scheduled workflow or a permission boundary.

Activity evidence

Lab	Minutes	Evidence of learning
1.1	10	Each application has a result and an owner for unresolved issues. Word output preserves 48 and 32 m3/h and does not diagnose pump failure.
1.2	15	Facts match the notes and unknowns stay unknown. Timing includes human review and the brief has a decision owner.

Facilitation and contingencies

Use the matching Facilitator Guide to time each lab, demonstrate the critical mechanism and review the expected outputs. Record work completed, work needing correction and unavailable live capabilities separately. Approved paired access supports participation; it does not prove individual deployment or permission tests. Never extend teaching into the protected lunch or tea breaks.

Module 2 Prompting for engineering documents

10:00-11:15 | 75 minutes | LO2, LO3

Slide reference: 20-42. Use the deck selectively for explanation and as a lab reference; do not lecture through every slide.

RISE and EMAP

R.I.S.E. means Role, Input, Steps and Expectations. It describes the writing task, names permissible evidence, specifies a reasoning or checking sequence and states the output criteria. E-MAP means Engineering, Metrics, Assumptions and Parameters. It makes the engineering context, measurable acceptance, unknowns and operating boundaries explicit. Use R.I.S.E. for a source-to-document task and E-MAP when uncertainty or constraints are central. Both require a verification pass.

Requirements and traceability

A defensible requirement includes an identifier, value, unit, source version, source location and acceptance evidence. In DES-01 revision 2, nominal delivery is 50 m3/h at 30 m head. The teaching tolerance is plus or minus 5 percent, giving a band of 47.5 to 52.5 m3/h. These are synthetic design-note values, not statutory or industry requirements. Wetted material, supply voltage and ambient conditions are deliberately unresolved; the correct draft preserves those gaps.

Activity evidence

Lab	Minutes	Evidence of learning
2.1	20	Teaching band is 47.5-52.5 m3/h. Every requirement cites its source or is marked missing. No invented material, voltage or approval.
2.2	15	All four timestamps and measurements are preserved. The liquid identity and calibration status remain unresolved.
2.3	20	All seven supplied hazard categories are addressed. Missing SDS and isolation information block release. Risk controls are proposals and not approved instructions.

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Module 3 Engineering methodologies and document automation

11:30-13:00 | 90 minutes | LO1, LO2, LO3

Slide reference: 43-77. Use the deck selectively for explanation and as a lab reference; do not lecture through every slide.

FMEA from function to action

Failure Mode and Effects Analysis examines how a function could fail, what effect follows, what could cause it and which current controls exist. Keep a failure mode distinct from its effect: no flow is a mode, underdose is an effect. The provided scale uses provisional severity, occurrence and detection ratings from 1 to 10; higher detection scores mean poorer detection. F01 gives 8 x 4 x 6 = 192; F02 gives 9 x 3 x 7 = 189; F03 gives 7 x 5 x 5 = 175. High-severity rows need attention even when their RPN is not the largest. A proposed action cannot justify reduced residual ratings before implementation and effectiveness evidence.

DMAIC connects a problem to a control plan

Define identifies the problem, scope and critical-to-quality characteristic. Measure establishes units, data quality and baseline. Analyse distinguishes signals and causal hypotheses from demonstrated causes. Improve proposes a controlled trial, owner and review gate. Control defines the metric, cadence, reaction plan and responsible person. A Pareto chart ranks counts; a fishbone organises possible causes; a 5-Whys chain requires evidence at each link. None of these tools alone establishes causation.

Activity evidence

Lab	Minutes	Evidence of learning
3.1	30	RPNs are 192, 189 and 175; complete rows remain aligned after sorting. Ratings have stated justifications and high severity is reviewed. Summary remains a draft awaiting review.
3.2	25	Baseline n=30, monitoring n=5; sample 33 signals above UCL. Cp and Cpk use the stated within-process estimate and the 9.4-10.6 L teaching limits. Pareto top two categories total 66.7%; causation is unproven.
3.3	20	All four components are present and consistent. Domain-specific gaps are retained and irrelevant pump criteria are excluded.

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Module 4 Copilot agents and reasoning agents

14:00-15:15 | 75 minutes | LO4, LO5

Slide reference: 78-101. Use the deck selectively for explanation and as a lab reference; do not lecture through every slide.

Agents make a task reusable

An agent combines a purpose, instructions, knowledge sources and reusable starting prompts. It does not remove the need for review. The FMEA Assistant must ask for missing rating scales, preserve the difference between controls and proposals, and return sources with provisional ratings. The Report Writer must follow the selected weekly or DMAIC template and preserve open actions. Grounding the agent on an obsolete or incomplete template can repeat the same mistake on every run.

Build and test the contract

In Microsoft 365 Copilot, use New agent and Configure or the available equivalent. Add a descriptive name, task instructions, selected training knowledge and starter prompts. Use Try it to test a normal request, missing source, conflicting versions and an embedded instruction to bypass review. Record observed responses, not only intended behaviour. Create or save the agent and share only within approved policy. A shared agent does not automatically grant the recipient access to its source files.

Activity evidence

Lab	Minutes	Evidence of learning
4.1	25	Normal and missing-source tests are recorded. Ratings and multiplication are checked independently. Sharing scope and source permissions are recorded.
4.2	20	Mandatory headings match the selected template. A02 is open and no unsupported percentage is generated.
4.3	15	Four unique actions and A04 has TBD owner and date. The report retains the approval blocker. Interactive execution is distinguished from an unattended scheduled flow.
4.4	15	At least two citations are opened and checked. Sample 33 is detected and baseline statistics match the reference.

Facilitation and contingencies

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Module 5 SharePoint integration with Copilot

15:30-16:15 | 45 minutes | LO6

Slide reference: 102-116. Use the deck selectively for explanation and as a lab reference; do not lecture through every slide.

Controlled knowledge for grounded answers

A SharePoint agent should use a deliberately selected library or set of current references. The teaching library contains a current design note, an obsolete revision, O&M information and a past-project summary. The current design note supports duty values; the past-project summary does not establish the cause of the current fault. Good retrieval depends on source scope, readable content, versions and document ownership.

Verify citations and conflicts

For each critical claim, record the document title, revision, clause or page, supported statement and source URL when available. Open the original source. DES-01 revision 1 says 45 m3/h but is explicitly obsolete; revision 2 says 50 m3/h. The answer should use the current reference and explain the conflict. Where materials remain TBD, the agent should say that evidence is missing rather than supply a likely grade.

Activity evidence

Lab	Minutes	Evidence of learning
5.1	25	Current revision is used and three factual claims are traceable. Missing material remains TBD. Permission test has actual evidence or is explicitly not executed.
5.2	15	Each claim can be traced through the handoff to the source. No uncertainty or permissions are lost during transfer.

Facilitation and contingencies

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Module 6 AI security and the engineering toolkit

16:15-16:45 | 30 minutes | LO7

Slide reference: 117-132. Use the deck selectively for explanation and as a lab reference; do not lecture through every slide.

Seven controls with evidence

The framework in the supplied proposal comprises data classification; data governance and access control; secure prompting; output verification and hallucination detection; human-in-the-loop sign-off; traceability and audit; and security awareness and incident response. Each control needs evidence or an explicit gap. The classroom classification categories are examples and must be mapped to the organisation policy before workplace use.

Verify before release

The seeded draft contains unsupported diagnosis, an obsolete duty, invented material compatibility, incorrect RPN, unjustified severity reduction, a false stability claim, invented certificate completion and false approval. A polished format does not mitigate these issues. The reviewer must check sources, calculations, units and unresolved assumptions. Record preparer, competent reviewer, evidence checked, release decision and reviewed version; the AI response itself cannot grant authority.

Activity evidence

Lab	Minutes	Evidence of learning
6.1	20	All seven control rows have evidence or an explicit gap. RPN error, obsolete revision, unsupported material and false approval are detected. Incident response preserves evidence and follows the approved route.
6.2	10	Two prompt frameworks and one testable next action are present. Live agent status is honest and a human review gate remains.

Facilitation and contingencies

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Alignment and completion evidence

Outcome	Primary evidence
LO1	Primer explanations; Lab 3.1 FMEA; Lab 3.2 chart and DMAIC interpretation
LO2	Labs 1.1, 1.2, 2.1-2.3 and 3.1-3.3 reviewed document outputs
LO3	RISE and E-MAP prompts, source checks and revision logs
LO4	Labs 4.1-4.3 configurations, actual tests and reviewed handoff
LO5	Lab 4.4 verified citations and independent statistical comparison
LO6	Labs 5.1-5.2 citation checks, access test and provenance handoff
LO7	Labs 6.1-6.2 seven controls, sign-off gate, incident record and toolkit

This package contains formative checks and a peer mini challenge. It does not add a certification exam or infer a formal assessment instrument absent from the supplied documents. Record any unexecuted outcome and agree follow-up through the provider delivery process.

Source authority: GenAI for Engineers proposal revised v2 PUB Feedback and Lesson Plan with Hands-On Labs v2 PUB Feedback. The source proposal trainer-profile placeholder is intentionally not copied; no trainer identity is invented.